

Joint Estimation Of State and Parameters for a Class of Nonlinear Disturbed Systems with Unknown Noise

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Abstract

This paper presents a Physics-Informed Bank of Estimators (PIBE) framework for jointly estimating states, parameters, and disturbances for a class of nonlinear disturbed systems in lower-triangular form. First, we introduce a bank-of-estimators approach for recursive state, parameter, and disturbance estimation, where each cell uses upstream estimates and a global objective backpropagates through the preceding cells during joint fine-tuning. Subsequently, we extend the method to compactly supported measurement-noise laws, including truncated-Gaussian and non-Gaussian cases. For the non-Gaussian case, we combine Physics-Informed Neural Networks (PINNs) with constrained scalar Energy-Based Models (EBMs) to fit flexible pooled residual densities. We later address error propagation through cascaded approximators and derive a conditional upper bound on the estimation error. The effectiveness of the proposed method is demonstrated through extensive numerical simulations.

Key words: State and parameter estimation, Physics-informed neural networks, Nonlinear systems, Energy-based models, Non-Gaussian noise, Observer design

1 Introduction

Physical systems are commonly represented by nonlinear dynamical models that are subject to model disturbances [15]. In a wide range of applications, accessing the system's state is crucial for control design, monitoring, fault detection, and diagnosis (see [11,22,23] for more details). However, in practice, measuring states directly is not always feasible due to physical or technological constraints. When direct measurements are unavailable, estimation algorithms, referred to as observers, are developed. The control community has extensively studied this problem, resulting in a wide variety of observers tailored to different system structures and available measurements (see [4] for a comprehensive literature review). Examples include Kalman-like filters (KF, EKF, UKF) [10,9], sliding mode observers (SMO) [24],

high-gain observers for systems in triangular canonical form [14,8], and Kazantzis–Kravaris/Luenberger (KKL) observers [12,2] based on a nonlinear change of coordinates that conjugates the dynamics to a stable linear form. While observer and estimation design for linear systems is well established, developing observers for nonlinear systems remains challenging. Moreover, a key challenge for these model-based observers is their reliance on an accurate dynamic model, which can lead to reduced performance in the presence of uncertainties; therefore, one needs to design observers robust to uncertainties. In addition to the system state, many physical systems depend on unknown parameters that may be constant or time-varying. Therefore, classical observers have been extended into adaptive observers [5] that jointly estimate states and parameters.

Beyond model uncertainty and unknown parameters, a second challenge in state and parameter estimation arises from the nature of the available measurements. In real-world scenarios, measurements are often noisy, and many proposed algorithms assume Gaussian noise profiles. However, this assumption can be too simplis-

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tic in many domains, such as bioengineering, geology, and economics, where non-Gaussian noise is prevalent. To address the limitations highlighted above, learning-based estimation algorithms have recently emerged, providing new approaches to state estimation through Neural Networks (NNs) that complement classical observer and parameter estimation techniques. Two main directions have been explored: a learning algorithm can itself serve as an estimator, or it can be used as a solver for a standard observer. For example, in [18], the authors propose learning the mapping of the nonlinear Kazantzis–Kravaris/Luenberger observer, an approach later extended to time-varying output-delay systems in [17]. Similarly, a Recurrent Neural Network (RNN)-based nonlinear observer that combines RNNs with the Extended Luenberger Observer (ELO) to effectively capture nonlinear system dynamics is introduced in [1]. These studies, among others, have paved the way for learning-based observer design, enabling new strategies for solving the observation and estimation problem. However, treating learning algorithms as estimators introduces two important limitations: the need for large amounts of training data and the lack of formal convergence guarantees, which motivate the design of estimators that exploit structural knowledge to reduce data requirements and provide quantitative error bounds.

In this paper, we propose a learning-based method for jointly estimating the state, parameters, and disturbance of nonlinear dynamical systems in a specific observability canonical form. Leveraging the Physics-Informed Neural Network (PINN) framework, we embed the system’s ordinary differential equation (ODE) residual into the training loss, thereby reducing data requirements while exploiting the structural information already used by classical observers. The designed chain of estimators then achieves accurate state and parameter estimation for disturbed nonlinear systems while simultaneously estimating unknown disturbances. Inspired by [21], we also integrate PINNs with constrained scalar EBM so that compactly supported non-Gaussian residual densities can be fitted without prescribing a named parametric family. This flexibility still requires explicit support and regularity assumptions and does not, by itself, guarantee robustness or identifiability. A preliminary version of this work was presented at the 2026 European Control Conference [6], where the PIBE framework was introduced with an error-propagation bound limited to state estimation under absent or Gaussian measurement noise. The present paper extends that version in three directions: it introduces the Energy-Based PIBE (EPIBE) for non-Gaussian measurement-noise scenarios, generalizes the error analysis to the joint estimation of states, parameters, and disturbance, and validates the framework on a higher-order nonlinear system including non-Gaussian scenarios. The main contributions of this work are: (i) we develop a PINN-based bank of estimators (PIBE) capable of jointly estimating states, parameters, and disturbance

for nonlinear systems in observability canonical form; (ii) we extend it to an Energy-Based PINN framework (EPIBE) that jointly fits a constrained pooled residual-density model and the physical estimator; (iii) we derive a conditional upper bound on error propagation within the chain under the stated regularity, Lipschitz, and post-training approximation assumptions. The paper is organized as follows: Section 2 describes the class of nonlinear systems considered. Section 3 details the proposed framework for the joint estimation of states, parameters, and disturbances under compactly supported truncated-Gaussian and non-Gaussian noise. Section 4 analyses the propagation of estimation error and derives a bound on the cumulated error. Section 5 presents numerical results, and Section ?? concludes the paper.

Notation. The sets of real and nonnegative real numbers are denoted by $\mathbb{R}$ and $\mathbb{R}_{\geq 0}$, respectively. The symbol I_q denotes the identity matrix of dimension q , and $\text{col}(v_1, \dots, v_m)$ denotes the column vector obtained by stacking its arguments. For a vector v , $\|v\|_2$ denotes the Euclidean norm. For $\mathbf{v} = [v_1, \dots, v_N]^\top \in \mathbb{R}^N$, we define the normalized discrete L^2 norm by $\|\mathbf{v}\|_{N,2} := \left(\frac{1}{N} \sum_{i=1}^N |v_i|^2 \right)^{1/2}$. For a measurable signal $v : [0, T] \rightarrow \mathbb{R}^m$, $\|v\|_{L^\infty(0,T)} := \text{ess sup}_{t \in [0,T]} \|v(t)\|_2$. When v is continuous, $\|v\|_\infty$ denotes the corresponding supremum norm. For a map g defined on a set $\mathcal{K}$, $\text{Lip}_{\mathcal{K}}(g)$ denotes its Lipschitz constant on $\mathcal{K}$. For symmetric matrices A and B , $A \succeq B$ means that $A - B$ is positive semidefinite. The notation $\mathbf{1}_{\mathcal{A}}$ denotes the indicator of a set $\mathcal{A}$, and “a.e.” means almost everywhere.

2 Preliminaries and problem formulation

We consider the following class of nonlinear systems subject to a time-varying disturbance in the last state, and measurement noise, written in triangular canonical form with multiple nonlinearities:

$$\begin{cases} \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} x_2 + f_1(x_1, \theta_1) \\ \vdots \\ x_n + f_{n-1}(x_1, \dots, x_{n-1}, \theta_1, \dots, \theta_{n-1}) \\ f_n(x, \theta) + d(t) \end{bmatrix}, \\ y = x_1 + \omega(t), \end{cases} \quad (1)$$

where $n \geq 2$ denotes the state dimension, $x = [x_1, \dots, x_n]^\top$ is the system’s state vector with $x(t) \in \mathbb{R}^n$, and $y(t) \in \mathbb{R}$ is the measured output signal. For $i = 1, \dots, n-1$, the known nonlinearity f_i has the signature $f_i(x_1, \dots, x_i, \theta_1, \dots, \theta_i)$, whereas the last known nonlinearity is $f_n(x, \theta)$. The unknown constant parameter vector is $\theta = (\theta_1, \dots, \theta_{n-1}) \in \Theta \subset \mathbb{R}^{n-1}$, where $\Theta = \Theta_1 \times \dots \times \Theta_{n-1}$ and every $\Theta_j \subset \mathbb{R}$ is nonempty and compact. On the finite estimation horizon $[0, T]$, the disturbance is assumed to admit the decomposition

$$d(t) = \Gamma_q(t)^\top a + r_q(t), \quad (2)$$

where $\Gamma_q(t) = [\gamma_0(t) \cdots \gamma_{q-1}(t)]^\top$ is a known vector of basis functions $\gamma_i \in C^1([0, T])$, $i = 0, \dots, q-1$, $a \in \mathcal{A} \subset \mathbb{R}^q$ is an unknown constant coefficient vector, and $\mathcal{A}$ is compact. The basis functions are assumed to be linearly independent on $[0, T]$, equivalently,

$$W_\Gamma := \int_0^T \Gamma_q(t) \Gamma_q(t)^\top dt \succeq \gamma_d I_q, \quad \gamma_d > 0. \quad (3)$$

The term $r_q \in L^\infty([0, T])$ accounts for the disturbance component not represented by the selected basis and satisfies

$$\|r_q\|_{L^\infty(0, T)} \leq \varepsilon_{d, q}. \quad (4)$$

The exact finite-dimensional disturbance class is recovered when $\varepsilon_{d, q} = 0$. Since $\mathcal{A}$ is compact and Γ_q is continuous on $[0, T]$, the nominal part $\Gamma_q^\top a$ is uniformly bounded; hence $d \in L^\infty([0, T])$. Finally, ω denotes a measurement-noise process whose realizations satisfy $\|\omega\|_{L^\infty(0, T)} \leq \bar{\omega}$.

The triangular structure in (1) is the observability canonical form underlying the design of high-gain and cascade observers [8, 14]. This structure is restrictive, but it represents a broad subclass of single-output nonlinear systems that can be mapped locally into an observability normal form under the appropriate observability-rank and coordinate-regularity conditions [13].

Assumption 1 (*Finite-horizon well-posedness and compact admissibility*) The sets $\chi_0 \subseteq \chi \subset \mathbb{R}^n$ are compact. For every $x_0 \in \chi_0$, $\theta \in \Theta$, $a \in \mathcal{A}$, and every remainder r_q satisfying (4), System (1) has a unique solution $x(\cdot; x_0, \theta, a, r_q)$ on $[0, T]$, and this solution remains in χ . For every admissible noise realization, the corresponding measured output remains in a compact set $\mathcal{Y} \subset \mathbb{R}$ on $[0, T]$.

System (1) can be written in the following compact form:

$$\begin{cases} \dot{x}(t) = f(x(t), \theta) + E_d d(t), \\ y(t) = Cx(t) + \omega(t), \end{cases} \quad (5)$$

where $E_d = [0, \dots, 0, 1]^\top$ and $C = [1, 0, \dots, 0]$. Let $\mathcal{O} \subset \mathbb{R}^n \times \mathbb{R}^{n-1}$ be an open set containing $\chi \times \Theta$. We assume that $f \in C^1(\mathcal{O}; \mathbb{R}^n)$. Consequently, partial derivatives of f w.r.t its arguments exist and are bounded on the compact admissible set $\chi \times \Theta$.

Assumption 2 (*Joint finite-horizon identifiability and local conditioning*) Let $y_0(t) := Cx(t)$ denote the noise-free output and define the joint unknown vector and its admissible set by $\eta := (x_0, \theta, a)$, $\mathcal{P} := \chi_0 \times \Theta \times \mathcal{A}$. For the nominal finite-dimensional disturbance model $d(t) = \Gamma_q(t)^\top a$, corresponding to $r_q = 0$ in (2), define the finite-horizon output map

$$\begin{aligned} \mathcal{O}_T : \mathcal{P} &\longrightarrow L^2(0, T), \\ \eta &\longmapsto y_0(\cdot; \eta). \end{aligned} \quad (6)$$

The following properties hold:

(i) Structural identifiability: for every $\eta, \tilde{\eta} \in \mathcal{P}$,

$$\mathcal{O}_T(\eta) = \mathcal{O}_T(\tilde{\eta}) \text{ a.e. on } [0, T] \implies \eta = \tilde{\eta}. \quad (7)$$

(ii) Local conditioning: for the true vector $\eta^* = (x_0^*, \theta^*, a^*)$, there exist a neighborhood $\mathcal{N}(\eta^*)$ and a constant $c_{\text{id}} > 0$ such that

$$\|\mathcal{O}_T(\eta) - \mathcal{O}_T(\eta^*)\|_{L^2(0, T)} \geq c_{\text{id}} \|\eta - \eta^*\| \quad (8)$$

for every $\eta \in \mathcal{N}(\eta^*) \cap \mathcal{P}$. When $r_q \neq 0$, it is treated as the bounded model mismatch specified in (4); consequently, exact recovery is not asserted and the resulting bounds retain a term depending on $\varepsilon_{d, q}$. Condition (8) is a local inverse-Lipschitz observability condition. It is closely related to the uniform injectivity conditions used in nonlinear observer design and to the smallest-eigenvalue characterization of the local observability Gramian; see, e.g., [3, 16].

Having the system's output measurements and the known nonlinearities f_i , this study aims at estimating the states $x_k(t)$ for $k = 2, \dots, n$, together with the parameters $\theta = (\theta_1, \dots, \theta_{n-1})$ and the time-varying unknown disturbance $d(t)$. For this purpose, we propose a bank of PINN-based estimators for recursive estimation of states, parameters, and disturbance. In this context, a bank of estimators refers to a sequence of estimators cascaded together, each one taking as input the concatenated outputs of the previous ones.

3 Proposed PIBE and EPIBE frameworks

In this section, we propose a PINN-based approach to jointly estimate the state, parameters, and disturbance of the system described by Equation (1) from the measured output $y(t)$.

3.1 Proposed Physics-Informed Bank of Estimators

In this subsection, we present the PIBE framework for classical zero mean noise case, which also covers the noise-free setting. The PIBE consists of a cascade of cells indexed by $k = 2, \dots, n+1$. Cells $k = 2, \dots, n$ perform recursive state and parameter estimation, whereas the final cell $k = n+1$ performs disturbance estimation. A hatted quantity carries two indices: the subscript identifies the state or parameter coordinate, whereas the superscript identifies the cell that produced it. For each state coordinate, let $\mathcal{X}_j \subset \mathbb{R}$ be the compact interval containing the j -th coordinate projection of χ . Let $\ell \in \{1, \dots, P\}$ index a trajectory and let $\{t_i\}_{i=1}^N \subset [0, T]$ be the fixed data grid. Define the sampled measurement vector

$$\mathbf{Y}^\ell := [y^\ell(t_1) \cdots y^\ell(t_N)]^\top \in \mathcal{Y}^N \subset \mathbb{R}^N, \quad (9)$$

and, for each upstream state estimate,

$$\hat{\mathbf{X}}_j^{j, \ell} := [\hat{x}_j^{j, \ell}(t_1) \cdots \hat{x}_j^{j, \ell}(t_N)]^\top \in \mathcal{X}_j^N. \quad (10)$$

For $k = 2, \dots, n+1$, the trajectory-level input to the k -th cell is

$$\mathbf{U}_{k-1}^\ell := \text{col}(\mathbf{Y}^\ell, \hat{\mathbf{X}}_2^{2,\ell}, \dots, \hat{\mathbf{X}}_{k-1}^{k-1,\ell}, \hat{\theta}_1^{2,\ell}, \dots, \hat{\theta}_{k-2}^{k-1,\ell}) \in \mathcal{U}_{k-1}, \quad (11)$$

where empty sequences are omitted, so that $\mathbf{U}_1^\ell = \mathbf{Y}^\ell$. More precisely, where an empty Cartesian product is understood as the singleton containing the empty tuple, the admissible input set is:

$$\mathcal{U}_{k-1} := \mathcal{Y}^N \times \prod_{j=2}^{k-1} \mathcal{X}_j^N \times \prod_{j=1}^{k-2} \Theta_j \subset \mathbb{R}^{d_{k-1}}, \quad (12)$$

which is compact, where $d_{k-1} = N(k-1) + (k-2)$.

For every $k = 2, \dots, n$, the k -th PIBE cell consists of a trajectory encoder, a continuous-time state decoder, and a time-independent parameter head. First, the encoder computes

$$\mathbf{z}_{k-1}^\ell = \mathcal{E}_k(\mathbf{U}_{k-1}^\ell; \Theta_{E_k}) \in \mathbb{R}^{r_k}. \quad (13)$$

The state decoder is the map

$$\mathcal{S}_k : [0, T] \times \mathbb{R}^{r_k} \longrightarrow \mathcal{X}_{k-1} \times \mathcal{X}_k, \quad (14)$$

$$(t, \mathbf{z}_{k-1}^\ell) \longmapsto \begin{bmatrix} \hat{x}_{k-1}^{k,\ell}(t) \\ \hat{x}_k^{k,\ell}(t) \end{bmatrix},$$

whereas the parameter head satisfies

$$\begin{aligned} \mathcal{Q}_k : \mathbb{R}^{r_k} &\longrightarrow \Theta_{k-1}, \\ \mathbf{z}_{k-1}^\ell &\longmapsto \hat{\theta}_{k-1}^{k,\ell}. \end{aligned} \quad (15)$$

Consequently, the complete trajectory-conditioned PINN is

$$\begin{aligned} \Phi^k : [0, T] \times \mathcal{U}_{k-1} &\longrightarrow (\mathcal{X}_{k-1} \times \mathcal{X}_k) \times \Theta_{k-1}, \\ (t, \mathbf{U}_{k-1}^\ell) &\longmapsto (\mathcal{S}_k(t, \mathbf{z}_{k-1}^\ell), \mathcal{Q}_k(\mathbf{z}_{k-1}^\ell)), \end{aligned} \quad (16)$$

with parameter block $\Theta_{PINN}^k := (\Theta_{E_k}, \Theta_{S_k}, \Theta_{Q_k})$. Since the parameter head does not receive t , $\hat{\theta}_{k-1}^{k,\ell}$ is constant over $[0, T]$ by construction. The state decoder uses twice continuously differentiable activation functions. Its first time derivative,

$$\dot{\hat{x}}_{k-1}^{k,\ell}(t) = \frac{\partial}{\partial t} \mathcal{S}_{k,1}(t, \mathbf{z}_{k-1}^\ell) \quad (17)$$

is computed by Automatic Differentiation (AD) while $\mathbf{z}_{k-1}^\ell$ is held fixed. By convention, $\hat{x}_1^2 \equiv \hat{x}_1$ denotes the reconstruction of the measured output. We write $\hat{x}_k := \hat{x}_k^k$ and $\hat{\theta}_{k-1} := \hat{\theta}_{k-1}^k$ when no ambiguity arises. Thus, the reported state trajectory is $\hat{\mathbf{x}} := (\hat{x}_1^2, \hat{x}_2^2, \dots, \hat{x}_n^n)$. The final cell also reconstructs $\hat{x}_n^{n+1}$, an auxiliary trajectory used in the last physics residual and in the disturbance-coefficient estimate. For readability, the trajectory index ℓ is suppressed in the loss expressions below; the reported training loss is the empirical average of these trajectory-wise losses over $\ell = 1, \dots, P$.

Because $\mathbf{U}_{k-1}^\ell$ contains measurements sampled over the complete interval $[0, T]$, the resulting PIBE is a finite-horizon estimator and is not causal in time. A causal

version would replace $\mathbf{U}_{k-1}^\ell$ by a past-only sliding window; that extension is outside the scope of this work.

Training proceeds sequentially, with two optimization phases for each newly added cell. During local pre-training of cell k , the upstream cells are frozen and their outputs are detached from the computational graph; only Θ_{PINN}^k is optimized by minimizing the cellwise loss $\mathcal{L}_{Loc}^k$, which combines a data-fit or inter-cell consistency term with a physics term. During joint fine-tuning, all cells in the currently constructed prefix $2:k$ are optimized using the weighted loss $\mathcal{L}_{Tot}^k$, without detaching the upstream outputs, so that gradients propagate through the cascade. Thus, “local” and “global” refer to the optimization scope: the current cell or the complete constructed prefix, respectively. For $k = n+1$, the global loss covers the entire bank. The successive cells and their losses are detailed below.

-Step 1: for $k = 2$, the sampled measurement trajectory $\mathbf{U}_1 = \mathbf{Y}$ is passed through $\mathcal{E}_2$. The decoder $\mathcal{S}_2$ reconstructs $(\hat{x}_1^2(t), \hat{x}_2^2(t))$, while the time-independent head $\mathcal{Q}_2$ estimates the constant parameter $\hat{\theta}_1^2$. The parameter block Θ_{PINN}^2 is optimized using the objective

$$\begin{aligned} \mathcal{L}_{Loc}^2(\Theta_{PINN}^2) &= \mathcal{L}_{Data}^2(y, \hat{x}_1^2 | \Theta_{PINN}^2) \\ &\quad + \lambda \mathcal{L}_{Physics}^2(\hat{x}_1^2, \hat{x}_2^2, \hat{\theta}_1^2 | \Theta_{PINN}^2), \end{aligned} \quad (18)$$

where the data term measures the mismatch between the measurement and the PINN reconstruction,

$$\mathcal{L}_{Data}^2(y, \hat{x}_1^2 | \Theta_{PINN}^2) = \frac{1}{N} \sum_{i=1}^N (y(t_i) - \hat{x}_1^2(t_i))^2, \quad (19)$$

and the physics term is the squared residual of the first ODE $\dot{x}_1 = x_2 + f_1(x_1, \theta_1)$ evaluated on the PINN outputs:

$$\mathcal{L}_{Physics}^2(\hat{x}_1^2, \hat{x}_2^2, \hat{\theta}_1^2 | \Theta_{PINN}^2) = \frac{1}{N_r} \sum_{j=1}^{N_r} |r_2(\tau_j)|^2, \quad (20)$$

with the residual

$$r_2(t) := \dot{\hat{x}}_1^2(t) - \hat{x}_2^2(t) - f_1(\hat{x}_1^2(t), \hat{\theta}_1^2). \quad (21)$$

By construction, r_2 is a function of the $k = 2$ PINN outputs and the known nonlinearity f_1 .

-Step 2: for $k = 3, \dots, n$, the encoder receives the trajectory-level input $\mathbf{U}_{k-1}$ defined in (11). The corresponding continuous upstream arguments are grouped as

$$\begin{aligned} \mathcal{I}_{k-2}^x(t) &:= \text{col}(\hat{x}_1^2(t), \{\hat{x}_j^j(t)\}_{j=2}^{k-2}), \\ \mathcal{I}_{k-2}^\theta &:= \text{col}(\{\hat{\theta}_j^{j+1}\}_{j=1}^{k-2}), \\ \mathcal{I}_{k-1}(t) &:= (\mathcal{I}_{k-2}^x(t), \mathcal{I}_{k-2}^\theta). \end{aligned} \quad (22)$$

Empty indexed sequences are omitted. Thus, $\mathcal{I}_{k-2}^x$ contains the upstream state arguments $x_1, \dots, x_{k-2}$, while $\mathcal{I}_{k-2}^\theta$ contains $\theta_1, \dots, \theta_{k-2}$. These quantities are held fixed during local pre-training and are used to evaluate

the physics residual. The k -th cell estimates $x_k(t)$ and the constant parameter θ_{k-1} . The local loss $\mathcal{L}_{Loc}^k$ is associated with the model estimating (x_k, θ_{k-1}) and defined by

$$\mathcal{L}_{Loc}^k(\Theta_{PINN}^k) = \mathcal{L}_{Data}^k(\hat{x}_{k-1}^{k-1}, \hat{x}_{k-1}^k | \Theta_{PINN}^k) + \lambda \mathcal{L}_{Physics}^k(\mathcal{I}_{k-1}, \hat{x}_{k-1}^k, \hat{x}_k^k, \hat{\theta}_{k-1}^k | \Theta_{PINN}^k), \quad (23)$$

with the consistency-with-upstream data term

$$\mathcal{L}_{Data}^k(\hat{x}_{k-1}^{k-1}, \hat{x}_{k-1}^k | \Theta_{PINN}^k) = \frac{1}{N} \sum_{i=1}^N (\hat{x}_{k-1}^{k-1}(t_i) - \hat{x}_{k-1}^k(t_i))^2, \quad (24)$$

and the physics term given by the squared residual of the $(k-1)$ -th ODE,

$$\mathcal{L}_{Physics}^k(\cdots | \Theta_{PINN}^k) = \frac{1}{N_r} \sum_{j=1}^{N_r} |r_k(\tau_j)|^2, \quad (25)$$

where the residual is

$$r_k(t) := \hat{x}_{k-1}^k(t) - \hat{x}_k^k(t) - f_{k-1}(\mathcal{I}_{k-2}^x(t), \hat{x}_{k-1}^k(t), \mathcal{I}_{k-2}^\theta, \hat{\theta}_{k-1}^k). \quad (26)$$

Thus, r_k depends only on (i) the frozen upstream estimates in $\mathcal{I}_{k-1}$ and (ii) the current PINN outputs $\hat{x}_{k-1}^k$, $\hat{x}_k^k$, and $\hat{\theta}_{k-1}^k$. After local pre-training of the k -th cell, all cells in the constructed prefix are jointly fine-tuned using the weighted global loss

$$\mathcal{L}_{Tot}^k(\mathfrak{W}_{2:k}) = \sum_{i=2}^k e^{-\frac{k-i}{k}} \mathcal{L}_{Loc}^i(\Theta_{PINN}^i), \quad (27)$$

where $\mathfrak{W}_{2:k} := \{\Theta_{PINN}^i\}_{i=2}^k$ collects all PINN parameters up to step k . Comparing the two estimators' outputs through $\mathcal{L}_{Data}^k$ enforces consistency at the x_{k-1} coordinate and enables fine-tuning of upstream weights once gradients are propagated through the global loss (27).

-Step 3: After estimating (x_k, θ_{k-1}) for $k = 2, \dots, n$, we introduce a final cell, indexed by $n+1$, to estimate the disturbance coefficients. For trajectory ℓ , its encoder first computes

$$\mathbf{z}_n^\ell = \mathcal{E}_{n+1}(\mathbf{U}_n^\ell; \Theta_{E_{n+1}}) \in \mathbb{R}^{r_{n+1}}, \quad (28)$$

where $\mathbf{U}_n^\ell$ contains the measurement trajectory, all upstream state estimates, and all parameter estimates, according to (11). The final state decoder and coefficient head are defined by

$$\mathcal{S}_{n+1} : [0, T] \times \mathbb{R}^{r_{n+1}} \longrightarrow \mathcal{X}_n, \quad (t, \mathbf{z}_n^\ell) \longmapsto \hat{x}_n^{n+1, \ell}(t), \quad (29)$$

and

$$\mathcal{Q}_{n+1} : \mathbb{R}^{r_{n+1}} \longrightarrow \mathcal{A}, \quad \mathbf{z}_n^\ell \longmapsto \hat{a}^\ell. \quad (30)$$

The output of $\mathcal{Q}_{n+1}$ is constrained to the admissible set $\mathcal{A}$, for instance through an output reparameterization when $\mathcal{A}$ is a box. The complete final cell is therefore

$$\Phi^{n+1} : [0, T] \times \mathcal{U}_n \longrightarrow \mathcal{X}_n \times \mathcal{A}, \quad (t, \mathbf{U}_n^\ell) \longmapsto (\mathcal{S}_{n+1}(t, \mathbf{z}_n^\ell), \mathcal{Q}_{n+1}(\mathbf{z}_n^\ell)), \quad (31)$$

with $\Theta_{PINN}^{n+1} := (\Theta_{E_{n+1}}, \Theta_{S_{n+1}}, \Theta_{Q_{n+1}})$. Moreover,

$$\dot{\hat{x}}_n^{n+1, \ell}(t) = \frac{\partial}{\partial t} \mathcal{S}_{n+1}(t, \mathbf{z}_n^\ell) \quad (32)$$

is computed by AD while $\mathbf{z}_n^\ell$ is held fixed. Thus, $\hat{a}^\ell$ may differ between trajectories but is constant with respect to t within each trajectory. Define

$$\hat{\mathbf{x}}^{n+1}(t) := \text{col}(\hat{x}_1^2(t), \{\hat{x}_j^j(t)\}_{j=2}^{n-1}, \hat{x}_n^{n+1}(t)).$$

The middle sequence is empty when $n = 2$. This is the auxiliary state vector used only to evaluate the final-cell physics residual. Define also

$$\hat{\boldsymbol{\theta}} := (\hat{\theta}_1^2, \dots, \hat{\theta}_{n-1}^n).$$

The local loss of the final cell is

$$\mathcal{L}_{Loc}^{n+1}(\Theta_{PINN}^{n+1}) = \mathcal{L}_{Data}^{n+1} + \lambda \mathcal{L}_{Physics}^{n+1}, \quad (33)$$

where the consistency term is

$$\mathcal{L}_{Data}^{n+1} = \frac{1}{N} \sum_{i=1}^N |\hat{x}_n^n(t_i) - \hat{x}_n^{n+1}(t_i)|^2, \quad (34)$$

and the physics term associated with the last ODE is

$$\mathcal{L}_{Physics}^{n+1} = \frac{1}{N_r} \sum_{j=1}^{N_r} |r_{n+1}(\tau_j)|^2, \quad (35)$$

with

$$r_{n+1}(t) := \hat{x}_n^{n+1}(t) - f_n(\hat{\mathbf{x}}^{n+1}(t), \hat{\boldsymbol{\theta}}) - \Gamma_q(t)^\top \hat{a}. \quad (36)$$

The resulting disturbance estimate is

$$\hat{d}(t) := \Gamma_q(t)^\top \hat{a}. \quad (37)$$

Thus, $\hat{d}$ estimates the structured component of (2), while r_q represents the irreducible finite-basis approximation error. After local pre-training, all cells are jointly fine-tuned using the extended global loss

$$\mathcal{L}_{Tot}^{n+1} = \sum_{i=2}^{n+1} e^{-\frac{n+1-i}{n+1}} \mathcal{L}_{Loc}^i. \quad (38)$$

3.2 Energy-based extension for non-Gaussian measurement noise (EPIBE)

The quadratic data-fit and inter-cell consistency losses used in PIBE do not account for the shape of an unknown non-Gaussian residual distribution. We therefore extend PIBE to compactly supported non-Gaussian measurement noise, which may be asymmetric and have a nonzero mean. The resulting EPIBE preserves the bank architecture, trajectory-conditioned cells, physics losses, and recursive training strategy of PIBE. For each cell, it introduces a constrained scalar Energy-Based Model (EBM) [21] to represent the distribution of the corresponding data-fit or inter-cell residual, and replaces the associated quadratic loss with a negative log-likelihood. The following calculation motivates this replacement in the presence of nonzero-mean noise.

To motivate this modification, suppose, for illustration, that the measurement noise has time-independent moments $\mathbb{E}[\omega(t)] = \mu_\omega$ and $\text{Var}(\omega(t)) = \sigma_\omega^2$. For a fixed

candidate reconstruction $\hat{x}_1$, the population quadratic data risk is

$$\begin{aligned}\mathcal{R}_D(\hat{x}_1) &:= \frac{1}{T} \int_0^T \mathbb{E} \left[(y(t) - \hat{x}_1(t))^2 \right] dt \\ &= \sigma_\omega^2 + \frac{1}{T} \int_0^T (\hat{x}_1(t) - x_1(t) - \mu_\omega)^2 dt.\end{aligned}\quad (39)$$

Consequently, the quadratic data risk alone is minimized by $\hat{x}_1 = x_1 + \mu_\omega$, rather than by the physical trajectory x_1 , whenever $\mu_\omega \neq 0$. Moreover, a quadratic penalty does not characterize the shape of an asymmetric or otherwise non-Gaussian residual distribution. EPIBE therefore replaces the quadratic residual penalty with a learned negative log-likelihood. This replacement does not, by itself, separate the physical signal from the residual distribution: physical identification continues to rely on the model constraints and Assumption 2.

For every cell $k = 2, \dots, n+1$, define the scalar data-fit residual by

$$\varepsilon_k(t) := \begin{cases} y(t) - \hat{x}_1^2(t), & k = 2, \\ \hat{x}_{k-1}^{k-1}(t) - \hat{x}_{k-1}^k(t), & k = 3, \dots, n+1. \end{cases} \quad (40)$$

For $k = 2$, this residual coincides with the measurement noise only when the physical output is correctly reconstructed. For $k \geq 3$, it measures the discrepancy between two consecutive cells at their shared state coordinate and therefore also reflects propagated upstream errors. Since these downstream residuals need not follow the original measurement-noise distribution, a separate scalar EBM is assigned to each cell. Each EBM is fitted to residual samples pooled across the training trajectories and sampling times and consequently represents a pooled marginal density.

For each cell $k = 2, \dots, n+1$, select a fixed compact residual support $\mathcal{R}_k = [-\bar{\rho}_k, \bar{\rho}_k]$, where the radii satisfy $\bar{\rho}_2 \geq \max_{(v,s) \in \mathcal{Y} \times \mathcal{X}_1} |v-s|$, $\bar{\rho}_k \geq \max_{u,v \in \mathcal{X}_{k-1}} |u-v|$, $k = 3, \dots, n+1$. These maxima are finite by compactness. In view of (40), this construction ensures that every admissible residual belongs to its corresponding support $\mathcal{R}_k$. Let $E_{\zeta_k} : \mathcal{R}_k \rightarrow \mathbb{R}$ denote the scalar energy function of the k -th EBM. Since adding a constant to an energy does not change the represented density, define its reference-point normalization by

$$\tilde{E}_{\zeta_k}(\xi) := E_{\zeta_k}(\xi) - E_{\zeta_k}(0). \quad (41)$$

Assumption 3 (Admissible scalar EBM class)

For every $k = 2, \dots, n+1$, the EBM parameter set $\mathcal{K}_k$ is compact, and the map

$$\zeta_k \mapsto \tilde{E}_{\zeta_k}$$

is continuous from $\mathcal{K}_k$ into $C(\mathcal{R}_k)$, equipped with the uniform norm.

By Assumption 3 and compactness of $\mathcal{K}_k$, there exists a finite constant $B_{E,k}$ such that

$$\max_{\zeta_k \in \mathcal{K}_k} \left\| \tilde{E}_{\zeta_k} \right\|_{L^\infty(\mathcal{R}_k)} \leq B_{E,k}. \quad (42)$$

The compact parameter constraint can, for example, be enforced by projected parameter updates.

For a fixed inverse-temperature parameter $\beta > 0$, the residual density represented by the k -th EBM is defined as

$$\begin{aligned}p_{k,\zeta_k}(\xi) &:= \frac{\exp(-\beta \tilde{E}_{\zeta_k}(\xi))}{Z_k(\zeta_k)} \mathbf{1}_{\mathcal{R}_k}(\xi), \\ Z_k(\zeta_k) &:= \int_{\mathcal{R}_k} \exp(-\beta \tilde{E}_{\zeta_k}(u)) du.\end{aligned}\quad (43)$$

The uniform energy bound in (42) yields

$$\begin{aligned}2\bar{\rho}_k e^{-\beta B_{E,k}} &\leq Z_k(\zeta_k) \leq 2\bar{\rho}_k e^{\beta B_{E,k}}, \\ \frac{e^{-2\beta B_{E,k}}}{2\bar{\rho}_k} &\leq p_{k,\zeta_k}(\xi) \leq \frac{e^{2\beta B_{E,k}}}{2\bar{\rho}_k}, \quad \xi \in \mathcal{R}_k.\end{aligned}\quad (44)$$

Hence $Z_k(\zeta_k)$ is finite and strictly positive, so that p_{k,ζ_k} is a well-defined probability density on $\mathcal{R}_k$. Moreover, the density bounds prevent the admissible EBM from forming arbitrarily concentrated likelihood spikes.

Using the residuals defined in (40), the empirical energy-based data loss of cell k is the average negative log-likelihood

$$\begin{aligned}\mathcal{L}_{\text{Data}}^{\text{ebm},k} &:= -\frac{1}{PN} \sum_{\ell=1}^P \sum_{i=1}^N \log p_{k,\zeta_k}(\varepsilon_k^\ell(t_i)) \\ &= \log Z_k(\zeta_k) + \frac{\beta}{PN} \sum_{\ell=1}^P \sum_{i=1}^N \tilde{E}_{\zeta_k}(\varepsilon_k^\ell(t_i)),\end{aligned}\quad (45)$$

for $k = 2, \dots, n+1$. The second equality follows from (43) because every admissible residual belongs to $\mathcal{R}_k$. This loss depends on the PINN parameters through the residual samples and on the EBM parameters through ζ_k . During mini-batch training, the empirical average is evaluated over the selected trajectories.

Writing $\Theta_{EBM}^k := \zeta_k \in \mathcal{K}_k$, the unified local EPIBE objective is

$$\mathcal{L}_{\text{Loc},E}^k(\Theta_{PINN}^k, \Theta_{EBM}^k) := \mathcal{L}_{\text{Data}}^{\text{ebm},k} + \lambda \mathcal{L}_{\text{Physics}}^k, \quad (46)$$

for $k = 2, \dots, n+1$. Here, $\mathcal{L}_{\text{Physics}}^k$ is exactly the corresponding PIBE physics loss: (20) for $k = 2$, (25) for $k = 3, \dots, n$, and (35) for $k = n+1$. Thus, EPIBE replaces only the quadratic data-fit and inter-cell consistency terms with the energy-based negative log-likelihood (45). During local joint training, the upstream outputs are detached, and only Θ_{PINN}^k and Θ_{EBM}^k are updated.

After local joint training of the k -th cell, all PINN and EBM parameters in the constructed prefix are jointly fine-tuned using for $k = 2, \dots, n+1$:

$$\mathcal{L}_{\text{Tot},E}^k(\mathfrak{W}_{2:k}^E) := \sum_{m=2}^k e^{-\frac{k-m}{k}} \mathcal{L}_{\text{Loc},E}^m, \quad (47)$$

where $\mathfrak{W}_{2:k}^E := \{(\Theta_{PINN}^m, \Theta_{EBM}^m)\}_{m=2}^k$ collects all trainable parameters in the current prefix. During this fine-

tuning phase, upstream outputs are not detached, so gradients of $\mathcal{L}_{\text{Tot},E}^k$ propagate through all preceding PINN cells and their associated EBMs. For $k = n + 1$, (47) includes the entire stateparameterdisturbance estimation bank.

The cells are trained sequentially for $k = 2, \dots, n + 1$. For each cell, choose iteration indices satisfying $0 < N_{\text{EBM}} < N_{\text{par}} < N_{\text{tot}}$, and use the following three-phase schedule. During the first N_{EBM} iterations, the EBM is inactive, the upstream cells are frozen, and the current PINN is trained using the original PIPE local loss $\mathcal{L}_{\text{Loc}}^k$. During iterations $N_{\text{EBM}}, \dots, N_{\text{par}} - 1$, the upstream outputs remain detached, while Θ_{PINN}^k and Θ_{EBM}^k are jointly updated using $\mathcal{L}_{\text{Loc},E}^k$, with the EBM parameter update projected onto $\mathcal{K}_k$. Finally, during iterations $N_{\text{par}}, \dots, N_{\text{tot}} - 1$, all PINN and EBM parameters in the current prefix are jointly fine-tuned using $\mathcal{L}_{\text{Tot},E}^k$, without detaching the upstream outputs.

Since each partition function Z_k is a one-dimensional integral over the fixed interval $\mathcal{R}_k$, it is evaluated during training using a fixed Gauss–Legendre quadrature rule. Gradients are backpropagated through the resulting quadrature sum. The initial PINN-only phase prevents the EBM from being fitted to residuals generated by a completely untrained physical estimator.

Proposition 1 (*Conditional population interpretation of the EBM*) Fix a cell k and a specified pooling distribution over trajectories and sampling times. Suppose that the resulting scalar consistency residual has a pooled marginal density p_k^* supported on $\mathcal{R}_k$, with finite differential entropy. For a fixed, correctly identified physical signal, define the population EBM risk

$$\mathcal{R}_{E,k}(\zeta_k) := \mathbb{E}_{\xi \sim p_k^*} [-\log p_{k,\zeta_k}(\xi)].$$

Then

$$\mathcal{R}_{E,k}(\zeta_k) = h(p_k^*) + D_{\text{KL}}(p_k^* \| p_{k,\zeta_k}), \quad (48)$$

where $h(p_k^*)$ is independent of ζ_k . Consequently, if p_k^* belongs to the density class in (43), every population-risk minimizer represents p_k^* almost everywhere, although the minimizing network parameter need not be unique. If p_k^* is not in the class, any population-risk minimizer is instead a KL projection onto the admissible density class.

Proof 1 Add and subtract $-\mathbb{E}_{p_k^*}[\log p_k^*(\xi)]$ in the definition of $\mathcal{R}_{E,k}$. This yields (48). The conclusions follow from the nonnegativity of the Kullback–Leibler divergence and its equality condition. $\square$

Proposition 1 is conditional on a correctly identified physical signal and concerns the population risk. It does not establish consistency of the empirical joint PINN–EBM training, which would additionally require suitable sampling or mixing, generalization, and optimization guarantees. Moreover, since the EBM is unconditioned, it represents only the pooled marginal residual density,

rather than a time-, state-, or trajectory-conditioned law.

The unified PIPE–EPIPE training procedure is summarized in Algorithm 1 and illustrated in Figure 1.

4 Joint state, parameter, and disturbance error propagation analysis

We analyze how cellwise estimation errors propagate through the recursive inputs to the downstream state, parameter, and disturbance estimates.

4.1 Scaled norms and error decomposition

We analyze one trajectory ℓ at a time and suppress the index ℓ whenever no ambiguity arises. For each $\star \in y, x_2, \dots, x_n, \theta_1, \dots, \theta_{n-1}, a$, let $\mathcal{S}_\star$ denote the corresponding compact admissible set and define the reference scale $s_\star := \sup_{\xi \in \mathcal{S}_\star} \|\xi\|_2 > 0$. Set $s_{x_1} := s_y$. For the trajectory vector $\mathbf{U}_{k-1}$ in (11), define

$$\|\mathbf{U}_{k-1}\|_{\mathcal{U}_{k-1}}^2 := \frac{\|\mathbf{Y}\|_{N,2}^2}{s_y^2} + \sum_{j=2}^{k-1} \frac{\|\mathbf{X}_j\|_{N,2}^2}{s_{x_j}^2} + \sum_{j=1}^{k-2} \frac{|\theta_j|^2}{s_{\theta_j}^2}, \quad (49)$$

where empty sums are zero. Equivalently, $\|\mathbf{U}_{k-1}\|_{\mathcal{U}_{k-1}} = \|\mathbf{D}_{k-1}\mathbf{U}_{k-1}\|_2$, where $\mathbf{D}_{k-1}$ divides each sampled-trajectory block by $\sqrt{N}$ and its reference scale, and each parameter block by its reference scale. For $j = 2, \dots, n$, let $\mathbf{X}_j^\ell := [x_j^\ell(t_1) \cdots x_j^\ell(t_N)]^\top$ be the sampled true state. The exact input and the input generated recursively by the trained upstream cells are, respectively,

$$\begin{aligned} \mathbf{U}_{k-1}^{\star,\ell} &:= \text{col}(\mathbf{Y}^\ell, \mathbf{X}_2^\ell, \dots, \mathbf{X}_{k-1}^\ell, \theta_1, \dots, \theta_{k-2}), \\ \check{\mathbf{U}}_{k-1}^\ell &:= \text{col}(\mathbf{Y}^\ell, \check{\mathbf{X}}_2^{2,\ell}, \dots, \check{\mathbf{X}}_{k-1}^{k-1,\ell}, \check{\theta}_1^{2,\ell}, \dots, \check{\theta}_{k-2}^{k-1,\ell}). \end{aligned} \quad (50)$$

Here $\check{\mathbf{X}}_j^{j,\ell}$ contains the samples of $\check{x}_j^{j,\ell}$. Both inputs in (50) contain the same measured vector $\mathbf{Y}^\ell$; in particular, $\mathbf{U}_1^{\star,\ell} = \check{\mathbf{U}}_1^\ell = \mathbf{Y}^\ell$. For $k = 2, \dots, n + 1$, define the trained cell operator

$$\mathfrak{F}_k(\mathbf{U}) := \left(\mathcal{S}_k(\cdot, \mathcal{E}_k(\mathbf{U})), \mathcal{Q}_k(\mathcal{E}_k(\mathbf{U})) \right) \in \mathcal{Z}_k, \quad (51)$$

where $\mathcal{Z}_k := C([0, T]; \mathbb{R}^2) \times \mathbb{R}$ for $k = 2, \dots, n$, and $\mathcal{Z}_{n+1} := C([0, T]; \mathbb{R}) \times \mathbb{R}^q$. For $k = 2, \dots, n$, write $\mathbf{v} = (v_{k-1}, v_k)$ and define

$$\|\mathbf{v}\|_{\mathcal{S}_k} := \sup_{t \in [0, T]} \left(\frac{|v_{k-1}(t)|^2}{s_{x_{k-1}}^2} + \frac{|v_k(t)|^2}{s_{x_k}^2} \right)^{1/2}, \quad |\vartheta|_{\mathcal{Q}_k} := \frac{|\vartheta|}{s_{\theta_{k-1}}}. \quad (52)$$

For the final cell, set

$$\|v\|_{\mathcal{S}_{n+1}} := \frac{\|v\|_\infty}{s_{x_n}}, \quad |b|_{\mathcal{Q}_{n+1}} := \frac{|b|_2}{s_a}. \quad (53)$$

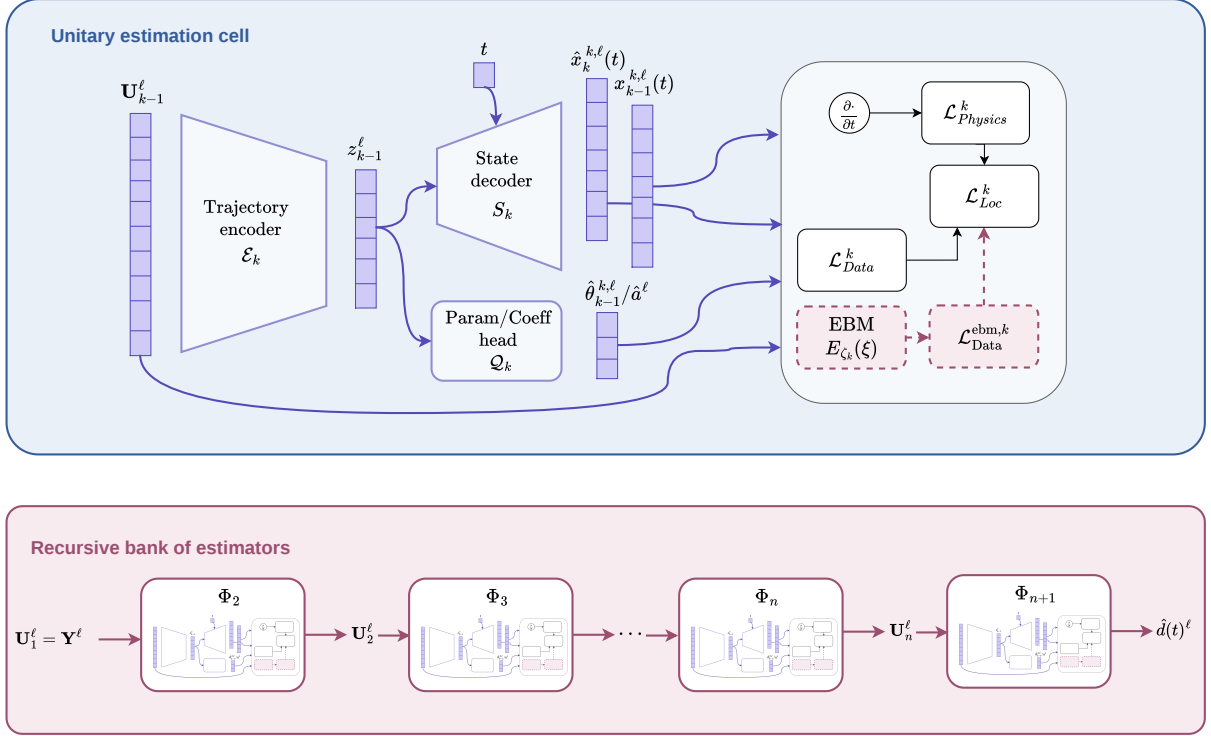


Fig. 1. Proposed estimation framework: **Top:** Offline training scheme. **Bottom:** Inference pipeline.

We equip the cell-output spaces with

$$\begin{aligned} \|(\mathbf{v}, \vartheta)\|_{\mathcal{Z}_k} &:= (\|\mathbf{v}\|_{\mathcal{Z}_k}^2 + \|\vartheta\|_{\mathcal{Z}_k}^2)^{1/2}, \quad k = 2, \dots, n, \\ \|(v, b)\|_{\mathcal{Z}_{n+1}} &:= (\|v\|_{\mathcal{Z}_{n+1}}^2 + \|b\|_{\mathcal{Z}_{n+1}}^2)^{1/2}. \end{aligned} \quad (54)$$

These dimensionless norms remove the physical units from the subsequent Lipschitz bounds. The exact-upstream and recursively generated cell outputs are

$$\bar{\mathbf{Z}}_k^\ell := \mathfrak{F}_k(\mathbf{U}_{k-1}^{\star,\ell}), \quad \check{\mathbf{Z}}_k^\ell := \mathfrak{F}_k(\check{\mathbf{U}}_{k-1}^\ell). \quad (55)$$

For $k = 2, \dots, n$, their new-state and parameter components are denoted $(\bar{x}_k^\ell, \bar{\theta}_{k-1}^\ell)$ and $(\check{x}_k^\ell, \check{\theta}_{k-1}^\ell)$, respectively. Their reconstructed-coordinate components are denoted $\bar{x}_{k-1}^{k,\ell}$ and $\check{x}_{k-1}^{k,\ell}$. For $k = n+1$, the corresponding components are $(\bar{x}_n^{n+1,\ell}, \bar{a}^\ell)$ and $(\check{x}_n^{n+1,\ell}, \check{a}^\ell)$.

For $k = 2, \dots, n$, define the intrinsic and propagated errors by

$$\begin{aligned} \epsilon_{s_k}^{\text{int}}(t) &:= \bar{x}_k(t) - x_k(t), \quad \epsilon_{p_{k-1}}^{\text{int}} := \bar{\theta}_{k-1} - \theta_{k-1}, \\ \epsilon_{s_k}^{\text{prop}}(t) &:= \check{x}_k(t) - \bar{x}_k(t), \quad \epsilon_{p_{k-1}}^{\text{prop}} := \check{\theta}_{k-1} - \bar{\theta}_{k-1}. \end{aligned} \quad (56)$$

Consequently,

$$\check{x}_k - x_k = \epsilon_{s_k}^{\text{prop}} + \epsilon_{s_k}^{\text{int}}, \quad \check{\theta}_{k-1} - \theta_{k-1} = \epsilon_{p_{k-1}}^{\text{prop}} + \epsilon_{p_{k-1}}^{\text{int}}. \quad (57)$$

When the superscript is omitted below, ϵ_{s_k} and $\epsilon_{p_{k-1}}$ refer to the corresponding intrinsic errors in (56). This decomposition separates the error generated by the current trained cell from that inherited through its trajectory-level input. For the final cell, define $\epsilon_a^{\text{int}} := \bar{a} - a$ and

$$\begin{aligned} \epsilon_a^{\text{prop}} &:= \check{a} - \bar{a}. \text{ Let } G_q := \sup_{t \in [0, T]} \|\Gamma_q(t)\|_2 < \infty, \\ \bar{d}_q(t) &:= \Gamma_q(t)^\top \bar{a}, \text{ and } \check{d}_q(t) := \Gamma_q(t)^\top \check{a}. \text{ Then} \\ \|d - \bar{d}_q\|_\infty &\leq G_q \|\epsilon_a^{\text{int}}\|_2 + \epsilon_{d,q}, \quad \|\check{d}_q - \bar{d}_q\|_\infty \leq G_q \|\epsilon_a^{\text{prop}}\|_2. \end{aligned}$$

For a feedforward neural network $\mathcal{N}$ with L layers, weight matrices $A^{(j)}$, and Lipschitz activation functions with constants α_j , the standard estimate $\text{Lip}(\mathcal{N}) \leq \prod_{j=1}^L \alpha_j \|A^{(j)}\|_2$ holds [19]. Assume that, on the admissible sets, the trained components satisfy

$$\begin{aligned} \|\mathcal{E}_k(\mathbf{U}) - \mathcal{E}_k(\tilde{\mathbf{U}})\|_2 &\leq L_{E_k} \|\mathbf{U} - \tilde{\mathbf{U}}\|_{\mathcal{Z}_{k-1}}, \\ \|\mathcal{S}_k(\cdot, z) - \mathcal{S}_k(\cdot, \tilde{z})\|_{\mathcal{S}_k} &\leq L_{S_k} \|z - \tilde{z}\|_2, \\ \|\mathcal{Q}_k(z) - \mathcal{Q}_k(\tilde{z})\|_{\mathcal{Q}_k} &\leq L_{Q_k} \|z - \tilde{z}\|_2. \end{aligned} \quad (58)$$

These constants can be bounded by applying the preceding estimate to the normalized maps induced by the input and output scalings. It follows from (54) that

$$\|\mathfrak{F}_k(\mathbf{U}) - \mathfrak{F}_k(\tilde{\mathbf{U}})\|_{\mathcal{Z}_k} \leq \Lambda_k \|\mathbf{U} - \tilde{\mathbf{U}}\|_{\mathcal{Z}_{k-1}}, \quad (59)$$

where

$$\Lambda_k := L_{E_k} \sqrt{L_{S_k}^2 + L_{Q_k}^2}, \quad k = 2, \dots, n+1. \quad (60)$$

When the cell constants are uniformly bounded, for example through spectral normalization or Lipschitz-constrained training [20, 7], define $\Lambda := \max_{2 \leq k \leq n+1} \Lambda_k$. The bound (59) is conditional on the trained network parameters and therefore applies to the PINN operators learned by either PIPE or EPIPE.

Algorithm 1 Unified training procedure for PIBE and EPIBE

Input: Mode $M \in \{\text{PIBE}, \text{EPIBE}\}$; sampled trajectories $\{y^\ell(t_i)\}_{\ell=1, i=1}^{P, N}$; system model (1); physics weight λ ; Per-cell iteration budget N_{tot} and local-training cutoff N_{par} , with $0 < N_{\text{par}} < N_{\text{tot}}$; For **EPIBE** only: $0 < N_{\text{EBM}} < N_{\text{par}}$, $\beta > 0$, residual supports $\{\mathcal{R}_k\}_{k=2}^{n+1}$, and constrained energy classes $\{\mathcal{K}_k\}_{k=2}^{n+1}$

Output: Trained PINN bank $\{\Phi^k\}_{k=2}^{n+1}$ and, for $M = \text{EPIBE}$, trained energy models

- 1: **for** $k \leftarrow 2$ **to** $n+1$ **do**
- 2: Initialize the current PINN parameters Θ_{PINN}^k
- 3: Freeze the upstream PINN cells $\{\Phi^m\}_{m=2}^{k-1}$
- 4: **if** $M = \text{PIBE}$ **then**
- 5: **for** $i \leftarrow 1$ **to** N_{par} **do**
- 6: Select a trajectory mini-batch
- 7: Construct $\mathbf{U}_{k-1}$ from detached upstream outputs, with $\mathbf{U}_1 = \mathbf{Y}$ for $k = 2$
- 8: Update only Θ_{PINN}^k by minimizing $\mathcal{L}_{\text{Loc}}^k$
- 9: **end for**
- 10: **else** ▷ EPIBE
- 11: Initialize $\Theta_{\text{EBM}}^k \in \mathcal{K}_k$
- 12: **for** $i \leftarrow 1$ **to** N_{EBM} **do**
- 13: Select a trajectory mini-batch
- 14: Construct $\mathbf{U}_{k-1}$ from detached upstream outputs, with $\mathbf{U}_1 = \mathbf{Y}$ for $k = 2$
- 15: Update only Θ_{PINN}^k by minimizing the PIBE loss $\mathcal{L}_{\text{Loc}}^k$
- 16: **end for**
- 17: **for** $i \leftarrow N_{\text{EBM}} + 1$ **to** N_{par} **do**
- 18: Select a trajectory mini-batch
- 19: Construct $\mathbf{U}_{k-1}$ from detached upstream outputs, with $\mathbf{U}_1 = \mathbf{Y}$ for $k = 2$
- 20: Jointly update $(\Theta_{\text{PINN}}^k, \Theta_{\text{EBM}}^k)$ by minimizing $\mathcal{L}_{\text{Loc}, E}^k$
- 21: Project Θ_{EBM}^k onto $\mathcal{K}_k$
- 22: **end for**
- 23: **end if**
- 24: Unfreeze all cells in the current prefix $\{\Phi^m\}_{m=2}^k$
- 25: **for** $i \leftarrow N_{\text{par}} + 1$ **to** N_{tot} **do**
- 26: Select a trajectory mini-batch
- 27: Evaluate $\{\Phi^m\}_{m=2}^k$ sequentially without detaching the upstream outputs
- 28: **if** $M = \text{PIBE}$ **then**
- 29: Update $\{\Theta_{\text{PINN}}^m\}_{m=2}^k$ by minimizing $\mathcal{L}_{\text{Tot}}^k$
- 30: **else**
- 31: Jointly update $\{\Theta_{\text{PINN}}^m, \Theta_{\text{EBM}}^m\}_{m=2}^k$ by minimizing $\mathcal{L}_{\text{Tot}, E}^k$
- 32: Project every Θ_{EBM}^m onto $\mathcal{K}_m$
- 33: **end if**
- 34: **end for**
- 35: **end for**

4.2 Conditional propagation through the estimator bank

For each admissible trajectory ℓ , define the ideal output of cell k by

$$\mathbf{Z}_k^{\star, \ell} := \begin{cases} (\text{col}(x_{k-1}^\ell(\cdot), x_k^\ell(\cdot)), \theta_{k-1}), & k = 2, \dots, n, \\ (x_n^\ell(\cdot), a^\ell), & k = n+1. \end{cases} \quad (61)$$

The uniform intrinsic error level of cell k is defined as

$$\bar{\varepsilon}_k^{\text{int}} := \sup_{\ell} \left\| \bar{\mathbf{Z}}_k^\ell - \mathbf{Z}_k^{\star, \ell} \right\|_{\mathcal{Z}_k}, \quad k = 2, \dots, n+1. \quad (62)$$

These quantities are finite because the target trajectories are bounded and the trained cell maps are continuous on the compact exact-input sets. No smallness condition is imposed at this stage. By the definitions of the scaled output norms, for $k = 2, \dots, n$:

$$\begin{aligned} \|\bar{x}_k^\ell - x_k^\ell\|_\infty &\leq s_{x_k} \bar{\varepsilon}_k^{\text{int}}, & |\bar{\theta}_{k-1} - \theta_{k-1}| &\leq s_{\theta_{k-1}} \bar{\varepsilon}_k^{\text{int}}, \\ \|\bar{x}_n^{n+1, \ell} - x_n^\ell\|_\infty &\leq s_{x_n} \bar{\varepsilon}_{n+1}^{\text{int}}, & \|\bar{a}^\ell - a^\ell\|_2 &\leq s_a \bar{\varepsilon}_{n+1}^{\text{int}}. \end{aligned} \quad (63)$$

Consequently, the intrinsic disturbance error satisfies

$$\|d^\ell - \bar{d}_q^\ell\|_\infty \leq G_q s_a \bar{\varepsilon}_{n+1}^{\text{int}} + \varepsilon_{d, q}. \quad (64)$$

For $k = 1, \dots, n$, define the upstream input discrepancy by

$$D_k^\ell := \left\| \check{\mathbf{U}}_k^\ell - \mathbf{U}_k^{\star, \ell} \right\|_{\mathcal{U}_k}, \quad (65)$$

with, $D_1^\ell = 0$. For $k = 2, \dots, n+1$, define the propagated and total cell errors by

$$\begin{aligned} P_k^\ell &:= \left\| \check{\mathbf{Z}}_k^\ell - \bar{\mathbf{Z}}_k^\ell \right\|_{\mathcal{Z}_k}, \\ E_k^\ell &:= \left\| \check{\mathbf{Z}}_k^\ell - \mathbf{Z}_k^{\star, \ell} \right\|_{\mathcal{Z}_k}. \end{aligned} \quad (66)$$

The cell Lipschitz bound (59) and the triangle inequality yield

$$\begin{aligned} P_k^\ell &\leq \Lambda_k D_{k-1}^\ell, \\ E_k^\ell &\leq \bar{\varepsilon}_k^{\text{int}} + \Lambda_k D_{k-1}^\ell, \end{aligned} \quad k = 2, \dots, n+1. \quad (67)$$

For $k = 2, \dots, n$, the input to the next cell retains the upstream blocks and appends the new-state samples and parameter estimate. Since the normalized sampled norm is bounded by the trajectory supremum norm, the scaled input and output norms give

$$(D_k^\ell)^2 \leq (D_{k-1}^\ell)^2 + (E_k^\ell)^2. \quad (68)$$

Consequently, for $k = 2, \dots, n$:

$$D_k^\ell \leq D_{k-1}^\ell + E_k^\ell \leq (1 + \Lambda_k) D_{k-1}^\ell + \bar{\varepsilon}_k^{\text{int}}. \quad (69)$$

Proposition 2 (Propagation through the estimator bank)

Under the cell Lipschitz bounds (59), for every admissible trajectory ℓ and every $m = 2, \dots, n+1$, the upstream discrepancy satisfies

$$D_{m-1}^\ell \leq \sum_{k=2}^{m-1} \bar{\varepsilon}_k^{\text{int}} \prod_{j=k+1}^{m-1} (1 + \Lambda_j). \quad (70)$$

The propagated and total cell errors therefore satisfy

$$P_m^\ell \leq \Lambda_m \sum_{k=2}^{m-1} \bar{\varepsilon}_k^{\text{int}} \prod_{j=k+1}^{m-1} (1 + \Lambda_j), \quad (71)$$

$$E_m^\ell \leq \bar{\varepsilon}_m^{\text{int}} + \Lambda_m \sum_{k=2}^{m-1} \bar{\varepsilon}_k^{\text{int}} \prod_{j=k+1}^{m-1} (1 + \Lambda_j). \quad (72)$$

Empty sums are zero, and empty products are one.

Proof 2 Starting from $D_1^\ell = 0$, induction in (69) gives (70). Substituting this bound into (67) yields (71) and (72), see ECC. $\square$

For $m = 2, \dots, n+1$, let $\mathcal{B}_m$ denote the right-hand side of (72).

Corollary 1 (Componentwise total-error bounds)

Under the hypotheses of Proposition 2, the reported state and parameter estimates satisfy

$$\|\check{x}_m^\ell - x_m^\ell\|_\infty \leq s_{x_m} \mathcal{B}_m, \quad (73)$$

$$|\check{\theta}_{m-1}^\ell - \theta_{m-1}| \leq s_{\theta_{m-1}} \mathcal{B}_m, \quad (74)$$

for $m = 2, \dots, n$. For the final cell,

$$\|\check{x}_n^{n+1,\ell} - x_n^\ell\|_\infty \leq s_{x_n} \mathcal{B}_{n+1}, \quad (75)$$

$$\|\check{a}^\ell - a^\ell\|_2 \leq s_a \mathcal{B}_{n+1}. \quad (76)$$

Consequently,

$$\|\check{d}_q^\ell - d^\ell\|_{L^\infty(0,T)} \leq G_q s_a \mathcal{B}_{n+1} + \varepsilon_{d,q}. \quad (77)$$

Proof 3 The state, parameter, and coefficient bounds follow from (72) by the coordinate projections of the scaled output norms. Since $\check{d}_q^\ell - d^\ell = \Gamma_q^\top (\check{a}^\ell - a^\ell) - r_q^\ell$ a.e. on $[0, T]$, the Cauchy-Schwarz inequality and (4) give $\|\check{d}_q^\ell - d^\ell\|_{L^\infty(0,T)} \leq G_q \|\check{a}^\ell - a^\ell\|_2 + \varepsilon_{d,q}$. Substitution of (76) proves (77). $\square$

4.3 Information provided by the local PIBE losses

We examine what the local quadratic losses control for one trajectory. For $k = 2, \dots, n$, consider the exact-upstream output $\bar{\mathbf{Z}}_k$ and define the sampled reconstruction residual by

$$g_{k,i}^\star := \begin{cases} \bar{x}_1^2(t_i) - y(t_i), & k = 2, \\ \bar{x}_{k-1}^k(t_i) - x_{k-1}(t_i), & k = 3, \dots, n. \end{cases} \quad (78)$$

The corresponding physics residual, with earlier state and parameter arguments set to their true values, is

$$r_k^\star(t) := \dot{\bar{x}}_{k-1}^k(t) - \bar{x}_k(t) - f_{k-1}(x_1(t), \dots, x_{k-2}(t), \bar{x}_{k-1}^k(t), \theta_1, \dots, \theta_{k-2}, \bar{\theta}_{k-1}), \quad (79)$$

where the earlier-state and earlier-parameter sequences are empty when $k = 2$. Define the exact-upstream quadratic losses by

$$\mathcal{L}_D^{k,\star} := \frac{1}{N} \sum_{i=1}^N |g_{k,i}^\star|^2, \quad \mathcal{L}_P^{k,\star} := \frac{1}{N_r} \sum_{j=1}^{N_r} |r_k^\star(\tau_j)|^2. \quad (80)$$

For $k = 2$, these coincide with the losses of the trained first cell. For $k \geq 3$, they involve true upstream signals and are generally unavailable from measurements alone.

For $k = 2, \dots, n$, define the reconstruction error

$$e_{k-1}^{\text{rec}}(t) := \bar{x}_{k-1}^k(t) - x_{k-1}(t). \quad (81)$$

Assumption 4 (Regularity for the local-loss estimates)

For each $k = 2, \dots, n$, $e_{k-1}^{\text{rec}} \in W^{2,\infty}(0, T)$ and $r_k^\star$ is Lipschitz continuous on $[0, T]$. There exist finite constants L_{g_k} , B_k , and L_{r_k} , independent of the trajectory and grid sizes under consideration, such that

$$\begin{aligned} \text{Lip}_{[0,T]}(e_{k-1}^{\text{rec}}) &\leq L_{g_k}, \\ \|e_{k-1}^{\text{rec}}\|_{L^\infty(0,T)} &\leq B_k, \\ \text{Lip}_{[0,T]}(r_k^\star) &\leq L_{r_k}. \end{aligned} \quad (82)$$

The ordered data and collocation grids satisfy $0 = t_1 < \dots < t_N = T$ and $0 = \tau_1 < \dots < \tau_{N_r} = T$. Their fill distances are

$$\begin{aligned} h_d &:= \sup_{t \in [0,T]} \min_{1 \leq i \leq N} |t - t_i|, \\ h_r &:= \sup_{t \in [0,T]} \min_{1 \leq j \leq N_r} |t - \tau_j|. \end{aligned} \quad (83)$$

Both grids are assumed quasi-uniform: for each grid, the ratio of the largest to the smallest adjacent spacing is bounded by a fixed constant $\kappa \geq 1$, independent of N and N_r .

Lemma 1 (From sampled losses to continuous-time bounds) Let $g : [0, T] \rightarrow \mathbb{R}$ be L -Lipschitz. Consider a grid $0 = s_1 < \dots < s_M = T$ with fill distance h and mesh ratio at most κ . Set $C_{\text{mesh}} := 4\kappa$. If

$$\frac{1}{M} \sum_{i=1}^M |g(s_i)|^2 \leq \varepsilon,$$

then

$$\frac{1}{T} \|g\|_{L^2(0,T)}^2 \leq C_{\text{mesh}} (\varepsilon + L^2 h^2), \quad (84)$$

and

$$\|g\|_\infty \leq \Psi_T(\varepsilon, h, L), \quad (85)$$

where

$$\Psi_T(\varepsilon, h, L) := \inf_{0 < \rho \leq T/2} \left\{ \sqrt{\frac{TC_{\text{mesh}}}{\rho} (\varepsilon + L^2 h^2)} + L\rho \right\}. \quad (86)$$

For fixed T , L , and κ , $\Psi_T(\varepsilon, h, L) \rightarrow 0$ as $\varepsilon, h \rightarrow 0$.

For $D, B \geq 0$, define

$$\mathcal{J}_T(D, B) := \inf_{0 < \rho \leq T/2} \left(\frac{2D}{\rho} + \frac{B\rho}{2} \right). \quad (87)$$

For $e \in W^{2,\infty}(0, T)$, identified with its C^1 representative, the bounds $\|e\|_\infty \leq D$ and $\|\dot{e}\|_{L^\infty(0,T)} \leq B$ imply

$$\|\dot{e}\|_\infty \leq \mathcal{J}_T(D, B). \quad (88)$$

Indeed, for any $t \in [0, T]$ and $0 < \rho \leq T/2$, choose $h \in \{-\rho, \rho\}$ such that $t + h \in [0, T]$. Taylor's formula

with integral remainder gives

$$|e(t+h) - e(t) - h\dot{e}(t)| \leq \frac{B\rho^2}{2}.$$

Consequently,

$$|\dot{e}(t)| \leq \frac{|e(t+h) - e(t)|}{\rho} + \frac{B\rho}{2} \leq \frac{2D}{\rho} + \frac{B\rho}{2}$$

. Taking the supremum over t and the infimum over ρ proves (88). For fixed B , $\mathcal{J}_T(D, B) \rightarrow 0$ as $D \rightarrow 0$.

For the first cell, assume additionally that $|\omega(t_i)| \leq \bar{\omega}$ for $i = 1, \dots, N$. Define

$$\varepsilon_{D,k}^\# := \begin{cases} 2\mathcal{L}_D^{2,*} + 2\bar{\omega}^2, & k = 2, \\ \mathcal{L}_D^{k,*}, & k = 3, \dots, n. \end{cases} \quad (89)$$

By (78),

$$\frac{1}{N} \sum_{i=1}^N |e_{k-1}^{\text{rec}}(t_i)|^2 \leq \varepsilon_{D,k}^\#. \quad (90)$$

For $k \geq 3$, this is an equality. For $k = 2$, it follows from $e_1^{\text{rec}}(t_i) = g_{2,i}^* + \omega(t_i)$ and $(a+b)^2 \leq 2a^2 + 2b^2$. Set

$$\begin{aligned} D_{0,k} &:= \Psi_T(\varepsilon_{D,k}^\#, h_d, L_{g_k}), \\ D_{1,k} &:= \mathcal{J}_T(D_{0,k}, B_k), \\ R_k &:= \Psi_T(\mathcal{L}_P^{k,*}, h_r, L_{r_k}). \end{aligned} \quad (91)$$

Under Assumption 4 and the stated grid conditions, these quantities satisfy

$$\begin{aligned} \|e_{k-1}^{\text{rec}}\|_\infty &\leq D_{0,k}, \\ \|\dot{e}_{k-1}^{\text{rec}}\|_\infty &\leq D_{1,k}, \\ \|r_k^*\|_\infty &\leq R_k. \end{aligned} \quad (92)$$

The first and third inequalities follow from Lemma 1, applied to e_{k-1}^{rec} and r_k^* , respectively. The second follows from (88).

For $k = 2, \dots, n$ and $s \in [0, 1]$, define

$$\begin{aligned} \xi_k(s, t) &:= \text{col}(x_1(t), \dots, x_{k-2}(t), x_{k-1}(t) + s e_{k-1}^{\text{rec}}(t), \\ &\quad \theta_1, \dots, \theta_{k-2}, \theta_{k-1} + s \epsilon_{p_{k-1}}^{\text{int}}), \end{aligned} \quad (93)$$

where empty sequences are omitted. Assume these segments lie in a fixed compact subset of the domain on which f_{k-1} is C^1 . Let $L_{f_{k-1}}^x$ bound $|\partial_{x_{k-1}} f_{k-1}|$ on this subset, and define

$$\begin{aligned} a_k(t) &:= \int_0^1 \partial_{x_{k-1}} f_{k-1}(\xi_k(s, t)) ds, \\ \beta_k(t) &:= \int_0^1 \partial_{\theta_{k-1}} f_{k-1}(\xi_k(s, t)) ds. \end{aligned} \quad (94)$$

In particular, $|a_k(t)| \leq L_{f_{k-1}}^x$. Subtracting the true $(k-1)$ -st state equation from (79) and applying the mean-value formula yields

$$\epsilon_{s_k}^{\text{int}}(t) + \beta_k(t) \epsilon_{p_{k-1}}^{\text{int}} = \dot{e}_{k-1}^{\text{rec}}(t) - a_k(t) e_{k-1}^{\text{rec}}(t) - r_k^*(t). \quad (95)$$

Consequently, (92) gives the coupled error bound

$$\left\| \epsilon_{s_k}^{\text{int}} + \beta_k \epsilon_{p_{k-1}}^{\text{int}} \right\|_\infty \leq \eta_k, \quad (96)$$

where

$$\eta_k := D_{1,k} + L_{f_{k-1}}^x D_{0,k} + R_k. \quad (97)$$

This inequality controls a combination of the new-state and parameter errors; separate bounds require additional information.

Corollary 2 (Conditional parameter-error bound)

Fix $k \in \{2, \dots, n\}$ and suppose (96) holds. Assume that an independent estimate provides $\|\epsilon_{s_k}^{\text{int}}\|_\infty \leq S_k$, and that the finite-horizon sensitivity satisfies

$$\frac{1}{T} \int_0^T |\beta_k(t)|^2 dt \geq \gamma_k > 0. \quad (98)$$

Then

$$|\epsilon_{p_{k-1}}^{\text{int}}| \leq \frac{\eta_k + S_k}{\sqrt{\gamma_k}}. \quad (99)$$

Proof 4 Since $\epsilon_{p_{k-1}}^{\text{int}}$ is constant in time,

$$\begin{aligned} \sqrt{\gamma_k} |\epsilon_{p_{k-1}}^{\text{int}}| &\leq \frac{\|\beta_k \epsilon_{p_{k-1}}^{\text{int}}\|_{L^2(0,T)}}{\sqrt{T}} \\ &\leq \frac{\|\epsilon_{s_k}^{\text{int}} + \beta_k \epsilon_{p_{k-1}}^{\text{int}}\|_{L^2(0,T)}}{\sqrt{T}} + \frac{\|\epsilon_{s_k}^{\text{int}}\|_{L^2(0,T)}}{\sqrt{T}} \\ &\leq \eta_k + S_k, \end{aligned}$$

which proves (99). $\square$

4.4 Weak-form bound for the disturbance coefficients

We derive a coefficient-error bound from the final-cell quadratic losses and the upstream error bounds. Integration by parts transfers the derivative of the reconstruction error onto a known modulating function. For one trajectory, define the final consistency residual

$$c_{n+1}(t) := \ddot{x}_n^{n+1}(t) - \ddot{x}_n^n(t). \quad (100)$$

Let $\check{r}_{n+1}$ denote the physics residual (36) evaluated using the trained bank's recursively generated outputs. The corresponding quadratic losses are

$$\begin{aligned} \mathcal{L}_D^{n+1, \text{rec}} &:= \frac{1}{N} \sum_{i=1}^N |c_{n+1}(t_i)|^2, \\ \mathcal{L}_P^{n+1, \text{rec}} &:= \frac{1}{N_r} \sum_{j=1}^{N_r} |\check{r}_{n+1}(\tau_j)|^2. \end{aligned} \quad (101)$$

These coincide with the trajectory-wise final PIBE loss components. For EPIBE, they can be evaluated as quadratic residual diagnostics. Choose the weight and vector modulating function

$$\varpi(t) := \frac{t(T-t)}{T^2}, \quad \psi_q(t) := \varpi(t) \Gamma_q(t). \quad (102)$$

Since $\Gamma_q \in C^1([0, T]; \mathbb{R}^q)$, ψ_q is continuously differentiable and satisfies $\psi_q(0) = \psi_q(T) = 0$. Define the weighted Gramian

$$\begin{aligned} W_{\Gamma, \varpi} &:= \int_0^T \varpi(t) \Gamma_q(t) \Gamma_q(t)^\top dt, \\ \gamma_{d, \varpi} &:= \lambda_{\min}(W_{\Gamma, \varpi}). \end{aligned} \quad (103)$$

The basis condition (3) and the strict positivity of ϖ on $(0, T)$ imply $\gamma_{d,\varpi} > 0$. Let $\check{x}^{n+1}$ and $\check{\theta}$ denote the recursive state and parameter arguments in (36). Define

$$e_n^F(t) := \check{x}_n^{n+1}(t) - x_n(t), \quad (104)$$

$$\Delta f_n(t) := f_n(\check{x}^{n+1}(t), \check{\theta}) - f_n(x(t), \theta).$$

Subtracting the true final-state equation from the recursive residual gives, almost everywhere,

$$\check{r}_{n+1} = \dot{e}_n^F - \Delta f_n - \Gamma_q^\top (\check{a} - a) + r_q. \quad (105)$$

The model assumptions and decoder regularity imply $e_n^F \in W^{1,\infty}(0, T)$. Multiplying (105) by ψ_q and integrating by parts, using $\psi_q(0) = \psi_q(T) = 0$, yields

$$\begin{aligned} W_{\Gamma,\varpi}(\check{a} - a) = & - \int_0^T \dot{\psi}_q(t) e_n^F(t) dt \\ & - \int_0^T \psi_q(t) \Delta f_n(t) dt + \int_0^T \psi_q(t) r_q(t) dt \\ & - \int_0^T \psi_q(t) \check{r}_{n+1}(t) dt. \end{aligned} \quad (106)$$

Define

$$C_{\psi,0} := \|\psi_q\|_{L^2(0,T;\mathbb{R}^q)}, \quad C_{\psi,1} := \|\dot{\psi}_q\|_{L^2(0,T;\mathbb{R}^q)}. \quad (107)$$

By the Cauchy-Schwarz inequality, $\|r_q\|_{L^2(0,T)} \leq \sqrt{T} \varepsilon_{d,q}$, and the positive definiteness of $W_{\Gamma,\varpi}$,

$$\begin{aligned} \|\check{a} - a\|_2 \leq & \frac{1}{\gamma_{d,\varpi}} \left[C_{\psi,1} \|e_n^F\|_{L^2(0,T)} + C_{\psi,0} \left(\|\Delta f_n\|_{L^2(0,T)} + \|\check{r}_{n+1}\|_{L^2(0,T)} + \sqrt{T} \varepsilon_{d,q} \right) \right]. \end{aligned} \quad (108)$$

Assume that c_{n+1} and $\check{r}_{n+1}$ are Lipschitz in time, with constants $L_{c,n+1}$ and $L_{\check{r},n+1}$, respectively. Define

$$\begin{aligned} A_{D,n+1} &:= \sqrt{TC_{\text{mesh}}(\mathcal{L}_D^{n+1,\text{rec}} + L_{c,n+1}^2 h_d^2)}, \\ A_{P,n+1} &:= \sqrt{TC_{\text{mesh}}(\mathcal{L}_P^{n+1,\text{rec}} + L_{\check{r},n+1}^2 h_r^2)}. \end{aligned} \quad (109)$$

Lemma 1 gives

$$\|c_{n+1}\|_{L^2(0,T)} \leq A_{D,n+1}, \quad \|\check{r}_{n+1}\|_{L^2(0,T)} \leq A_{P,n+1}. \quad (110)$$

Since $e_n^F = c_{n+1} + (\check{x}_n^n - x_n)$, Corollary 1 yields

$$\|e_n^F\|_{L^2(0,T)} \leq H_{n+1} := A_{D,n+1} + \sqrt{T} s_{x_n} \mathcal{B}_n. \quad (111)$$

Assume that the true and recursively evaluated state-parameter arguments lie in a fixed compact subset of $\mathcal{O}$. Choose finite nonnegative constants $L_{n,j}^x$ and $L_{n,j}^\theta$ such that

$$\begin{aligned} |f_n(\tilde{x}, \tilde{\theta}) - f_n(x, \theta)| \leq & \sum_{j=1}^n L_{n,j}^x |\tilde{x}_j - x_j| \\ & + \sum_{j=1}^{n-1} L_{n,j}^\theta |\tilde{\theta}_j - \theta_j| \end{aligned} \quad (112)$$

on this set. Define

$$\begin{aligned} F_{n+1} := & L_{n,n}^x H_{n+1} + \sqrt{T} \left(L_{n,1}^x s_{x_1} \mathcal{B}_2 + \sum_{j=2}^{n-1} L_{n,j}^x s_{x_j} \mathcal{B}_j \right. \\ & \left. + \sum_{j=1}^{n-1} L_{n,j}^\theta s_{\theta_j} \mathcal{B}_{j+1} \right). \end{aligned}$$

The reconstructed-coordinate component of the full-cell bound for cell 2 controls the x_1 error. Applying Corollary 1 to the remaining upstream components, and using $\|v\|_{L^2(0,T)} \leq \sqrt{T} \|v\|_\infty$, therefore gives

$$\|\Delta f_n\|_{L^2(0,T)} \leq F_{n+1}. \quad (113)$$

Proposition 3 (*Weak-form coefficient and disturbance bounds*) Suppose the cell Lipschitz bounds (59) hold for $k = 2, \dots, n$. Assume the stated grid conditions, the temporal Lipschitz bounds used in (109), and the model bound (112) hold. Define

$$\begin{aligned} C_{a,n+1} := & \frac{1}{\gamma_{d,\varpi}} \left[C_{\psi,1} H_{n+1} \right. \\ & \left. + C_{\psi,0} (F_{n+1} + A_{P,n+1} + \sqrt{T} \varepsilon_{d,q}) \right]. \end{aligned} \quad (114)$$

Then the recursively generated coefficient and disturbance estimates satisfy

$$\|\check{a} - a\|_2 \leq C_{a,n+1}, \quad (115)$$

and

$$\|\check{d}_q - d\|_{L^\infty(0,T)} \leq G_q C_{a,n+1} + \varepsilon_{d,q}. \quad (116)$$

Proof 5 Substituting (110), (111), and (113) into (108) proves (115). Moreover,

$$\check{d}_q - d = \Gamma_q^\top (\check{a} - a) - r_q \quad \text{a.e. on } [0, T].$$

Therefore,

$$\|\check{d}_q - d\|_{L^\infty(0,T)} \leq G_q \|\check{a} - a\|_2 + \varepsilon_{d,q},$$

which gives (116). $\square$

5 Numerical simulations

We consider the following third-order nonlinear system:

$$\begin{aligned} \dot{x}_1 &= x_2 - 0.35 \tanh(x_1) + \theta_1 (1.3 + 0.7 \cos(x_1)), \\ \dot{x}_2 &= x_3 - 0.3 \tanh(x_2) + 0.1 \sin(x_1) + \theta_2 (1.2 + 0.6 \cos(2x_2)), \\ \dot{x}_3 &= -0.648x_1 - 2.34x_2 + 0.15 \sin(x_1) + 0.8 \tanh(x_1 x_2) \\ &\quad - 2.7x_3 + \theta_1 \sin(x_2) + \theta_2 \tanh(x_3) + d(t), \\ y &= x_1 + \omega(t). \end{aligned} \quad (117)$$

The admissible state-coordinate intervals and initial-condition set are chosen as $\mathcal{X}_1 = [-2.5, 2.5]$, $\mathcal{X}_2 = [-0.8, 0.8]$, $\mathcal{X}_3 = [-0.8, 0.8]$, $\mathcal{X}_0 = [-0.8, 0.8] \times [-0.5, 0.5] \times [-0.35, 0.35]$. The admissible parameter set is $\Theta = [-0.4, 0.4]^2$, while the parameters used to generate the training trajectories are sampled uniformly from $\Theta_{\text{train}} = [-0.25, 0.25]^2$. The simulation horizon is $T = 20$. We first consider the exact finite-dimensional disturbance model, with $q = 3$ and $r_3 = 0$, given by

$$\begin{aligned} \Gamma_3(t) &= \begin{bmatrix} \sin(\Omega t) & \cos(\Omega t) & \sin(2 \cdot \Omega t) \end{bmatrix}^\top, \\ d(t) &= \Gamma_3(t)^\top a, \quad a \in \mathcal{A} = [-0.18, 0.18]^3. \end{aligned} \quad (118)$$

Trajectory generation and data sets. For each trajectory ℓ , the initial condition, parameters, and disturbance coefficients are independently sampled according

Table 1

Network architectures.

Cell	Encoder	Decoder	Head	Params.
2	201 → 512 → 512 → 128	129 → 256 → 256 → 256 → 2	128 → 128 → 128 → 1	630 275
3	403 → 512 → 512 → 128	129 → 256 → 256 → 256 → 2	128 → 128 → 128 → 1	733 699
4	605 → 512 → 512 → 128	129 → 256 → 256 → 256 → 1	128 → 128 → 128 → 3	837 124

Table 2

Normalized test errors (%) under the considered noise scenarios.

Noise	Method	E_{x_2}	E_{x_3}	E_{θ_1}	E_{θ_2}	E_d
None	PIBE	2.17	3.41	2.64	2.32	2.06
None	EPIBE	—	—	—	—	—
Gaussian	PIBE	2.44	3.58	2.66	2.34	5.4
Gaussian	EPIBE	—	—	—	—	—
Non-Gaussian	PIBE	—	—	—	—	—
Non-Gaussian	EPIBE	—	—	—	—	—

to $x_0^\ell \sim \text{Unif}(\chi_0)$, $\theta^\ell \sim \text{Unif}(\Theta_{\text{train}})$, $a^\ell \sim \text{Unif}(\mathcal{A})$. The system is integrated over $[0, T]$, and the resulting trajectories are sampled at $t_i = \frac{(i-1)T}{N-1}$, $i = 1, \dots, N$, with $N = 201$. The same uniform grid is used for the physics collocation points, so that $N_r = N$ and $\tau_i = t_i$. A total of 4000 trajectories are generated and partitioned, by trajectory, into 3000 training, 500 validation, and 500 test trajectories. The reference scales used for normalization are $s_y = s_{x_1} = 2.5$, $s_{x_2} = s_{x_3} = 0.8$, $s_{\theta_1} = s_{\theta_2} = 0.4$, $s_a = 0.12\sqrt{3}$.

Measurement-noise scenarios. Three measurement scenarios are considered. In the noise-free case, $\omega^\ell(t_i) = 0$. In the truncated-Gaussian case, $\omega^\ell(t_i) \sim \mathcal{N}(0, 0.03^2) \mid |\omega^\ell(t_i)| \leq 0.09$. This distribution is symmetric, compactly supported, and has zero mean. For the non-Gaussian case, let $Z_i^\ell \sim \text{Beta}(2, 5)$ and define $\omega^\ell(t_i) = 0.20Z_i^\ell - 0.027$. The resulting noise is asymmetric, supported on $[-0.027, 0.173]$, and satisfies $\mathbb{E}[\omega] = 0.0301$, $\text{Std}(\omega) = 0.0319$. Noise samples are generated independently across trajectories and sampling times. The same clean trajectories are used in all three scenarios, with only the measurement-noise realizations changed.

Network architecture. Each cell uses an MLP encoder with a 128-dimensional latent space, a time-conditioned state decoder, and a parameter/coefficient head. The architectures are given in Table 1. The decoder and head outputs are scaled to their corresponding admissible state, parameter, and coefficient intervals.

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A Proofs

Proof 6 (Proof of Lemma 1) *Partition $[0, T]$ into Voronoi intervals V_i associated with the grid nodes s_i . Quasi-uniformity gives $|V_i| \leq 2\kappa T/M$, while $|t - s_i| \leq h$ for $t \in V_i$. Hence*

$$|g(t)|^2 \leq 2|g(s_i)|^2 + 2L^2h^2, \quad t \in V_i.$$

Integrating over the intervals and summing yields

$$\frac{1}{T} \|g\|_{L^2(0,T)}^2 \leq 4\kappa\varepsilon + 2L^2h^2 \leq C_{\text{mesh}}(\varepsilon + L^2h^2),$$

which proves (84). For any $t \in [0, T]$ and $0 < \rho \leq T/2$, choose an interval $I_t \subset [0, T]$ of length ρ having t as an endpoint. Lipschitz continuity and the Cauchy–Schwarz inequality give

$$|g(t)| \leq \frac{1}{\rho} \int_{I_t} |g(s)| ds + L\rho \leq \frac{\|g\|_{L^2(0,T)}}{\sqrt{\rho}} + L\rho$$

. Using (84), taking the supremum over t , and then the infimum over ρ proves (85). The stated limit follows by first letting $\varepsilon, h \rightarrow 0$ for fixed $\rho > 0$, and then letting $\rho \rightarrow 0$. $\square$